

SHRUTI

Data Scientist | ML & AI Engineer | Clinical NLP | LLM Agents

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SUMMARY

Data Scientist with **3+ years** of production experience building **clinical NLP**, **LLM agent (LangGraph)**, and **deep learning** systems at scale. Sole owner of **distributed training & MLOps pipelines (Azure ML, AWS SageMaker, DeepSpeed)**, with hands-on expertise in **RAG, Claude API, transformers**, and end-to-end model deployment. Proven track record of driving **4x efficiency gains**, achieving **90-95% model accuracy**, and acting as technical SME for cross-functional teams.

TECHNICAL SKILLS

ML / DL: PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, Mamba (SSM), Transformers, LSTM, GLU-CNN

LLM & Agents: LangGraph, LangFuse, GPT APIs, RAG Pipelines, Claude API, Chroma, FAISS, MCP Servers

MLOps & Cloud: Azure ML, AWS SageMaker, GitHub CI/CD, Docker, Kubernetes, MLflow, DeepSpeed, Delta Lake

Languages: Python, SQL, Java | Spark, Hadoop, REST APIs | Jira, Confluence, Agile/Scrum

EXPERIENCE

Data Scientist | Optum (UnitedHealth Group) — Hyderabad Jan 2024 – Present

- **Sole production owner of distributed training & MLOps pipelines** — handling end-to-end deployment, incident response, and architectural decisions for **clinical NLP** systems across production workloads.
- **Technical SME** for clinical AI — conducted structured KT for 2–3 new hire data scientists and 4–5 external contractors from Fractal Analytics and Wipro via recorded sessions; served as go-to resolver for production blockers.
- Presented model performance and clinical AI insights to product stakeholders in bi-weekly reviews — directly influencing iteration priorities and product decisions.
- Engineered multi-label clinical NLP pipeline (**Mamba SSM**) for auto-assignment of **CPT & ICD-CM codes** from radiology reports — **90% accuracy**, significantly reducing manual coding effort at production scale.
- Boosted accuracy to **92%** via **Mamba + XGBoost + Ridge ensemble**; developed **GLU-CNN** for short-context clinical text achieving **95% accuracy** with lower memory footprint than transformer baselines.
- Cut model training time **4x** via **DeepSpeed + multi-GPU distributed training**; drove architectural enhancements improving pipeline throughput and reducing end-to-end processing time.
- Delivered full MLOps automation: **GitHub CI/CD**, model registration, automated validation, **Azure ML + SageMaker** deployment, retraining pipelines for drift management, and **MLflow** experiment tracking.
- Won **Runners-Up** at internal Optum Innovation Hackathon — built distributed CPU-based training & deployment solution reducing compute cost and pipeline time (team of 4).
- Authored Confluence production pipeline documentation; authored Python evaluation scripts storing model metrics as shared team reference; applied **Responsible AI** principles across all deployments.

Data Science Intern | Tredence Inc. — Remote Jul 2023 – Dec 2023

- Built customer churn prediction model (**Random Forest, 96% accuracy**) for banking client — full ETL, feature engineering, hyperparameter tuning, and stakeholder storytelling.

PROJECTS

- **RAG Pipeline with Claude API (Personal Project):** Built an end-to-end **Retrieval-Augmented Generation** system using **Claude API** for generation and **Chroma/FAISS** for vector storage and semantic search — covering document chunking, embedding generation, retrieval ranking, and context-aware response generation.
- **Bitcoin Price Prediction (Personal Project):** Developed a time-series forecasting model using **LSTM** on historical Bitcoin price data combined with technical indicators (moving averages, RSI, volatility) to predict short-term price trends.

ACHIEVEMENTS & CERTIFICATIONS

Runners-Up Internal Optum Hackathon • Won Epsilon Women in Tech Hackathon 2023 (coding round) • Kaggle ICR Medical Condition Prediction 2023 • 300+ problems LeetCode / HackerEarth / GFG / Codeforces

AWS Cloud Essentials (Simplilearn) • AI Dojo — Healthcare AI Agents using LangGraph (Optum/UHG) • RAG, MCP Server & Claude AI Agents (Coursera, 2025)

EDUCATION

B.Tech – Information Technology | Banasthali Vidyapith, Rajasthan | 2019–2023 | CGPA: 8.1/10